

Machine Learning Regression for the Econometrics Student

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Abstract

Inferential vs predictive notions of the same statistical process: linear regression.

1 Inference vs prediction in statistics

1.1 In econometrics we infer

- In econometrics, regression is the engine of **inference**.
- Inference, in this sense, means we use existing data to infer the *relationship* between some variables (e.g. x = income and y = SAT score).
- To be more precise, this relationship is a statistical distribution. Our goal is to model that distribution.
- When we do regression, we assume that there's a *true* distribution; a **data generating process**.
- When we do linear regression, we assume that true data generating process looks like a multivariate linear equation:

$$y = X\beta + \epsilon$$

- Our **object of interest** is the vector of weights: β
- Put differently, β is the vector of true relationships between our variables.

1.2 Econometricians care about (a) error and (b) bias

1.3 In ML we predict

- In machine learning, our

- 2 Linear regression is just projecting onto $\text{span}(\mathbf{X})$
- 3 Gradient descent for OLS
- 4 OLS is MLE under Gaussian noise
- 5 Logistic regression doesn't have closed form
- 6 Regularizing is biasing on purpose